

LTSpice IV: Ideal for SMPS Simulations

LTSpice IV is a free, high-performance Spice tool to virtually model, simulate and analyse electronic circuits, particularly switching regulators, before actually building them using electronic components

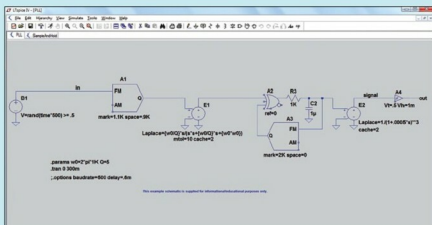
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LTSpice, formerly known as SwitcherCAD, is the fourth-generation free Spice simulation, schematic-capture and waveform-viewer software solution with enhancements and models offered by the semiconductor manufacturer Linear Technology Corporation (LTC).

This software features built-in component models to help obtain accurate results for non-linear designs. It also comes with various advanced and full-fledged capabilities that let you precisely test circuit functionalities. Exceptionally good simulation of switched mode power supply (SMPS) controllers and regulators is said to be one of the key features of this product.

Its speed of simulation has greatly improved as compared to the previous Spice programs, without compromising on the accuracy of simulation, due to the introduction of mixed-mode simulation capability and many other enhancements. In a few minutes it can perform detailed cycle-by-cycle SMPS simulations and analyse them.

It is to be noted that LTSpice is not SMPS-specific Spice. It has all the features of any other typical analogue and mixed-signal simulation program, with an added benefit that it is fast enough to simulate a SMPS interactively. This improved performance is an advantage for simulating general analogue circuits and would aid all electronics engineers.

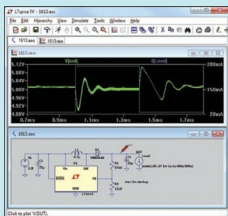


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Meant for whom?

LTSpice IV can be used to design simple and complex switching regulators and run simulations for these circuits. The ease to learn, simplicity of usage and extensive help files make it a good choice for students. The advanced simulation and analysis capabilities make it a great tool for creating professional-grade circuits by electronics engineers.

Many students, novice engineers and experienced professionals in such fields as radio-frequency electronics, power electronics, digital electronics and other disciplines use LTSpice IV. Those users who want to create their own schemes of integrated circuits and test them can also use this Spice circuit simulator. While it cannot generate printed circuit board (PCB) layouts, netlists can still be imported into layout programs.



Simulation

Stable and feature-rich

The software is highly stable, and the number of nodes you can create using LTSpice is unlimited. The application comes with a library that has a varied collection of predefined components like resistors, capacitors, inductors, diodes, wires, BUS taps, text boxes, labels etc that can be added to the cir-